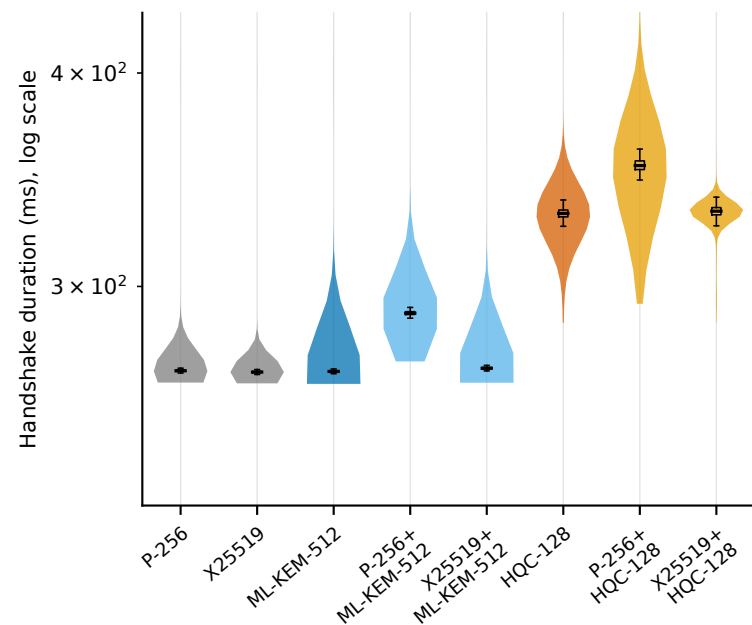
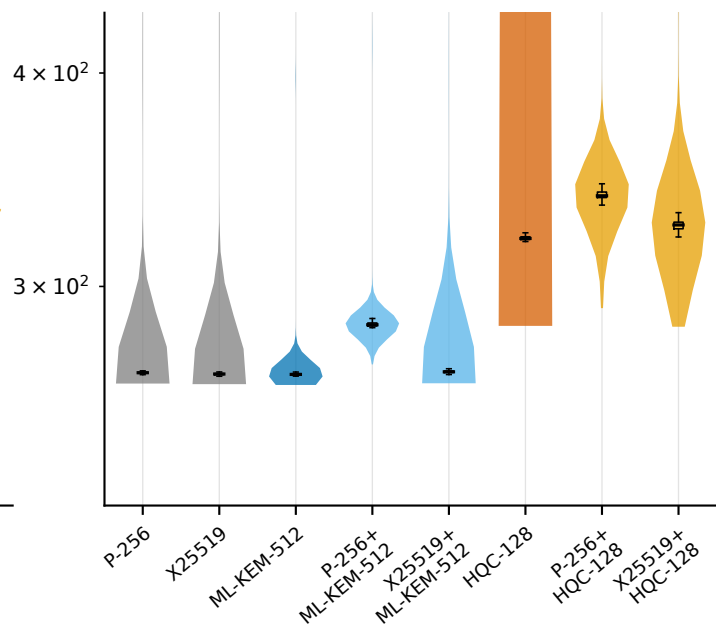


(a)

ed25519 (classical)

**(b)**

mldsa44 (post-quantum)

**(c)**

Classical KEX
 ML-KEM (pure)
 ML-KEM (hybrid)
 HQC (pure)
 HQC (hybrid)

KEM	ed25519 Mean (ms)	ed25519 CV (%)	ed25519 Max (ms)	mldsa44 Mean (ms)	mldsa44 CV (%)	mldsa44 Max (ms)
P-256	269.7	9.6	668.3	272.1	23.9	1.59k
X25519	268.4	6.9	661.8	272.6	24.7	1.53k
ML-KEM-512	272.2	20.4	1.29k	267.8	6.2	562.3
P-256+ML-KEM-512	292.0	18.2	1.47k	286.6	6.1	574.3
X25519+ML-KEM-512	271.9	18.7	1.33k	275.3	27.7	1.57k
HQC-128	335.2	13.4	829.1	1.00k	1513.9	339.28k
P-256+HQC-128	359.9	24.7	1.64k	342.3	14.9	1.34k
X25519+HQC-128	331.9	4.4	594.6	334.5	24.5	1.70k